ARUSHI SANJAY AGRAWAL

arushi.s.agrawal@gmail.com | linkedin.com/in/arushi-agrawal-4567b0174 | github.com/arushi-agrawal | +44 7721950269 | London, UK

SUMMARY

Machine Learning Engineer with 3+ years shipping production ML end-to-end. From research and model development through distributed pipelines to deployment at scale, with prior experience at large technology organisations. A computer vision and 3D perception specialist, currently one of two ML engineers at a 6-person startup, owning its entire defect-detection and point-cloud pipeline from model architecture to production serving.

EDUCATION

UNIVERSITY COLLEGE LONDON — London, UK
Master of Science, Machine Learning - Distinction — 2024 - 2025
- Thesis: Enhancing Spatial Reasoning in VLA models, Huawei
- Courses: Probabilistic and Unsupervised Learning, Bayesian Deep Learning, Artificial Intelligence for Sustainable Development, Applied ML, Applied DL, Statistical Natural Language Processing, Open-Endedness and General Intelligence, Computer Vision

INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI (IITG) — Guwahati, India
Bachelor of Technology, Engineering Physics, GPA - 8.45/10.0 — 2018 - 2022
- Thesis: Handling Bias in Visual Question Answering
- Courses: Linear Algebra, Multivariate Calculus, Statistics, Real and Complex Analysis, Computing, Mathematical Physics, Computational Physics, Probability & Statistical Mechanics, Pattern Recognition & Machine Learning

PROFESSIONAL EXPERIENCE

INFRAMIND LABS — London, UK
Machine Learning Engineer — Oct 2025 - Present
- Own end-to-end computer-vision and point-cloud ML pipeline for an automated tunnel-inspection product, detecting multiple defect types and structural attributes across 7+ client assets, from raw-data ingestion through distributed inference to client-ready platform outputs.
- Trained and iterated on multiple RGB-D segmentation models (Ultralytics YOLO, Mask2Former), authoring a custom recall-aware segmentation loss and tuning input/output representations to lift 3D defect recall by 27%.
- Orchestrated full pipeline and exposed it as a product platform (up to 88% faster end-to-end), with a FastAPI service layer and a web console for client teams to trigger and monitor runs.
- Authored a unified ML-systems design spec: layered datasets with a canonical 3D point cloud as source of truth, uniform sample-to-prediction contracts, a role-based model registry, tiered evaluation benchmarks, and semantic dataset and model versioning.
- Parallelised pipeline across an Azure Kubernetes (AKS) cluster, routing each workflow to its own node pool with horizontal pod autoscaling and per-stage CPU allocation, raising workflow parallelism from 0.97× to 13.5×.

HUAWEI — London, UK
Reinforcement Learning Intern — Apr 2025 - Sep 2025
- Developed a foundation model for scene graph generation from robotics video data, leveraging DINO, CLIP, SAM, and TAM for object detection, segmentation, and tracking.
- Fine-tuned Vision-Language Action (VLA) models with LoRA, strengthening spatial reasoning and action grounding.
- Integrated graph-based spatial representations into VLA architectures, achieving a 12% improvement on internal contextual reasoning benchmarks.
- Conducted robustness and generalisation analysis of VLA models (OpenVLA, π0) across multi-seed LIBERO manipulation rollouts in simulation, demonstrating overfitting as success rates collapsed from up to 100% to 0% under instruction and environment perturbations.

ADOBE — Bangalore, India
Software Development Engineer (GenAI Team) — Jul 2022 - Aug 2024
- Engineered and integrated multi-modal machine learning features into Adobe Premiere Pro (30M+ users), leveraging C++ and MVVM architecture to streamline video editing workflows, increasing user satisfaction.
- Optimised SegNeXt-based model for video mask generation using ONNX, improving accuracy by 17%.
- Analysed Hadoop-sourced user interaction data, driving a 13% increase in feature adoption through data-driven insights.

TEXT MERCATO SOLUTIONS — Remote
Machine Learning Intern — Sep 2021 - Mar 2022
- Implemented U2Net architecture for image segmentation and Boundary-Aware PoolNet for salient object detection.
- Enabled automated product description generation using Named Entity Recognition and LSTMs, reducing inference latency by 50%.

PROJECTS

RETRIEVAL-AUGMENTED LANGUAGE AGENTS FOR GAME PLAY Jan 2025 - Apr 2025
Open Endedness and AGI / Prof. Tim Rocktaschel, UCL GitHub, Paper

- Built a LLM-based AI agent on RAG and a FAISS vector database, adding a context-summarization step that lifted benchmark progression score 75% (0.4→0.7) over the unsummarized baseline.
- Trained a reward-ranking model with RLHF and a decision transformer for long-term action evaluation.

MEMORY-AUGMENTED CRITIQUE FOR SELF-IMPROVEMENT IN LLMS Jan 2025 - Apr 2025
Statistical Natural Language Processing / Prof. Pontus Saito Stenetorp GitHub, Paper

- Developed a multi-agent system with Super Tiny Language Models (77M parameters) with LoRA, prompt engineering, structured debate, and memory modules, reaching a ROUGE-1 F1 score of 0.406 in dialogue summarisation.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript/ TypeScript, HTML, CSS, MATLAB, R, SQL, LaTeX
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace
Libraries & Tools: NumPy, Pandas, OpenCV, Matplotlib, FAISS, CUDA, Git, MySQL, LangChain, Copilot, FastAPI, Docker, Kubernetes / AKS, Azure, CI/CD

LEADERSHIP & ACHIEVEMENTS

- **Advanced C++ Boot Camp, Adobe:** Ranked 1st by designing and implementing a high-performance application.
- **Shadowfax Cascade Cup 2022:** Secured 1st place out of 1,086 participants in a national Data Science Quiz.
- **Joint Entrance Exam (JEE) 2018:** Achieved 99.5 percentile among 1.26 million candidates in India.
- **Academic Mentor 2021-2022**: Guided team of 6 students in academics and career planning.